

Indian Institute of Technology Madras

Department of Electrical Engineering

Assignment 1

Simulation of Silicon Wafer: Resistance vs. Temperature

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Submitted By	
Name:	Shaun Joseph Alappatt
Roll Number:	EE26E007

Question 1: Intrinsic Silicon Characteristics

Problem Statement

Using a Verilog-A physical model, simulate an intrinsic (undoped) silicon wafer and provide the following three deliverables:

1. Plot Resistance vs. Temperature in a logarithmic y-scale at $V_d = 10$ V from 300 K to 600 K.
2. Plot Current vs. Temperature for $V_d = 10$ V.
3. Plot Current vs. Voltage (I-V) at different temperatures ranging from 300 K to 600 K.

Mathematical Model

Under the intrinsic (undoped) assumption, there are no intentional dopant atoms, meaning the donor concentration is zero ($N_D = 0$). The carrier concentration is solely determined by the intrinsic carrier concentration n_i, which is highly sensitive to temperature and governed by the standard relationship:

$$n_i = \sqrt{N_c(T)N_v(T)}\exp\left(-\frac{E_g}{2kT}\right)$$

As the temperature is swept from 300 K to 600 K, thermal energy exponentially excites electrons from the valence band across the bandgap (E_g) into the conduction band, causing n_i to increase by several orders of magnitude.

While the carrier mobility μ degrades due to increased phonon (lattice) scattering at higher temperatures:

$$\mu = U_0\left(\frac{T}{T_{ref}}\right)^{-1.5}$$

This reduction in mobility is completely overwhelmed by the massive exponential growth in free carriers. Consequently, the overall conductivity:

$$\sigma = q \cdot n_i \cdot \mu$$

increases drastically, meaning the resistance must plummet exponentially as temperature increases.

Verilog-A Implementation

```
`include "constants.vams"
`include "disciplines.vams"

module res_intrinsic(p, n);
    inout p, n;
    electrical p, n;

    parameter real len    = 10e-6    ;
    parameter real width  = 100e-6   ;
```

```
    parameter real thick   = 0.2e-6    ;
    parameter real Eg      = 1.12      ;
    parameter real Nc300   = 2.8e25    ;
    parameter real Nv300   = 1.02e25   ;
    parameter real U0      = 1350e-4   ;
    parameter real Temp    = 300.0     ; // Swept via schematic
    parameter real Tref    = 300.0     ;

    real Area, Nc, Nv, ni, mu, sigma, R;

    analog begin
        Area = width * thick;
        Nc = Nc300 * pow(Temp/Tref, 1.5);
        Nv = Nv300 * pow(Temp/Tref, 1.5);
        ni = sqrt(Nc * Nv) * exp(-(Eg * `P_Q) / (2.0 * `P_K * Temp));
        mu = U0 * pow(Temp/Tref, -1.5);
        sigma = ni * `P_Q * mu;
        R = len / (sigma * Area);
        V(p, n) <+ I(p, n) * R;
    end
endmodule
```

Listing 1: Verilog-A Code for Intrinsic Silicon Resistor

Circuit Schematic

To execute the DC temperature sweep, the Verilog-A component was placed in a testbench alongside a 10 V DC voltage source and a current probe. A parameter sweep block was configured to step the global `Temp` variable from 300 K to 600 K, which dynamically updates the temperature parameter inside the Verilog-A model during the simulation.

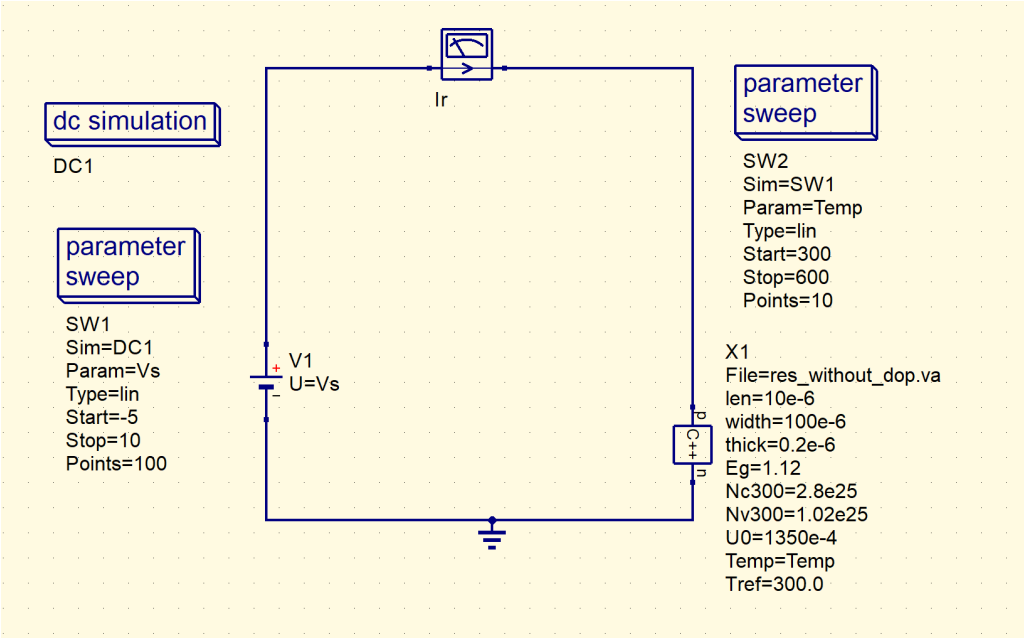


Figure 1: DC Simulation Testbench with Temperature Parameter Sweep

Results and Deliverables

1. Resistance vs. Temperature ($V_d = 10$ V)

A DC operating point simulation was executed using a 10 V DC source with a parameter sweep from 300 K to 600 K. As theoretically predicted, the intrinsic resistance plummeted by approximately five orders of magnitude due to the massive thermal generation of electron-hole pairs, requiring a logarithmic y-axis for proper visualization.

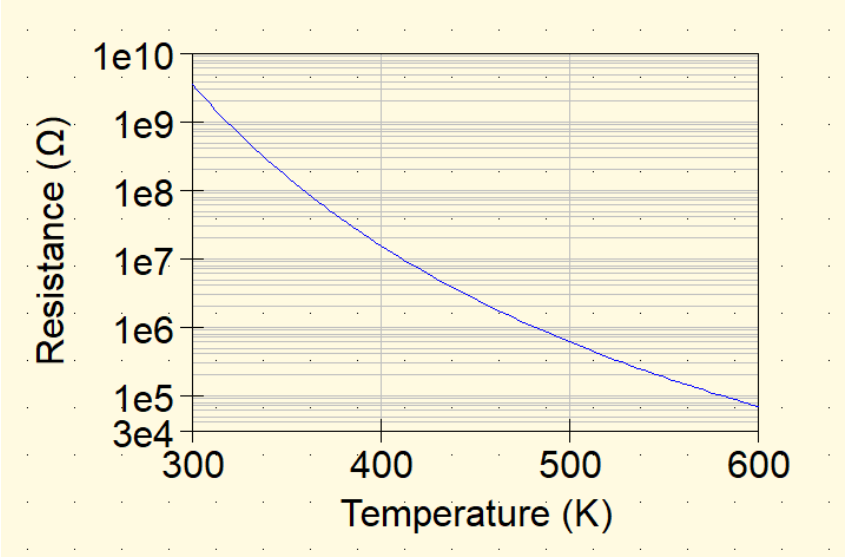


Figure 2: Resistance vs. Temperature for Intrinsic Silicon (Logarithmic Scale)

2. Current vs. Temperature ($V_d = 10$ V)

The current exhibits an exponential growth mirroring the resistance drop, confirming that additional thermal energy excites more electrons across the bandgap into the conduction band.

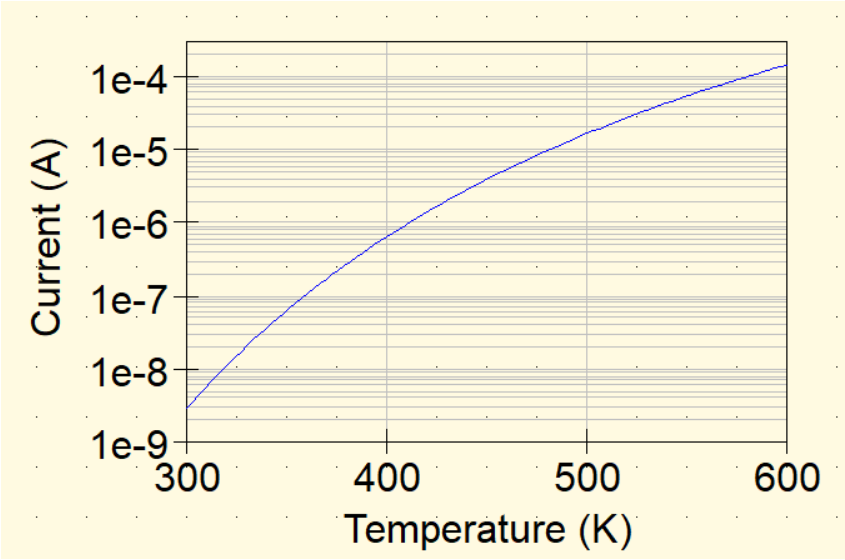


Figure 3: Current vs. Temperature for Intrinsic Silicon at $V_d = 10$ V

3. Current vs. Voltage (I-V) at Different Temperatures

By sweeping the voltage while stepping the temperature parameter from 300 K to 600 K, a family of I-V curves was generated. The slope of each curve represents the conductance $(1/R)$. As expected for a resistor, the curves remain linear (ohmic), but the slope becomes progressively steeper at higher temperatures due to the increased carrier concentration.

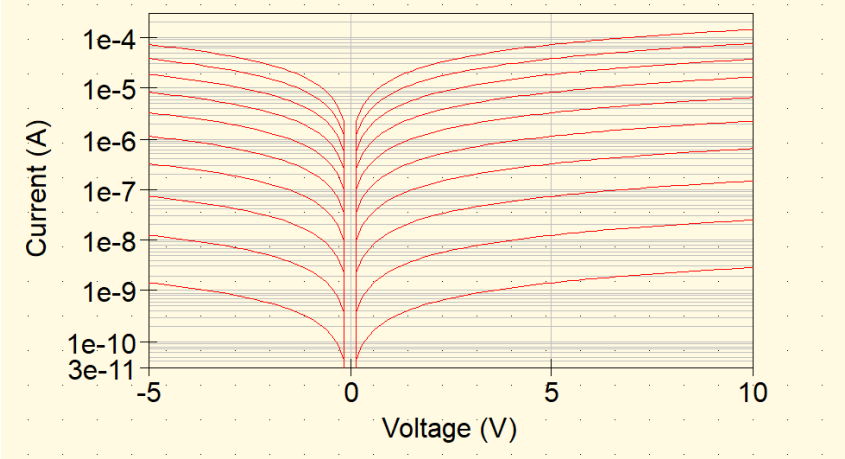


Figure 4: I-V Curves at Multiple Temperatures (300 K to 600 K)

Inference

The results of the simulation analysis for the intrinsic silicon wafer clearly bring out the overwhelming nature of thermal generation of charge carriers in semiconductors. In the intrinsic case, thermal energy causes excitation of electrons from the valence band to the conduction band across the bandgap. As the thermal generation of charge carriers is an exponential function of temperature, there is a sharp increase in the value of n_i between 300K and 600K. Due to this large number of charge carriers, the reduction in mobility of charge carriers due to scattering with the lattice becomes negligible. Thus, there is an exponential increase in the conductivity of the material and hence the resistance falls exponentially. The current increases exponentially as well. The I-V analysis demonstrates that even though the intrinsic silicon is ohmic (linear) for a particular static temperature, the conductance of the material increases sharply with increasing temperature of the surroundings.

Question 2: Resistance vs. Temperature for N-Doped Silicon

Problem Statement

Using a Verilog-A physical model, simulate an n-doped silicon wafer assuming full ionization of dopants with a constant donor concentration of $N_D = 2 \times 10^{23}$ m^{-3}, and provide the following three deliverables:

1. Plot Resistance vs. Temperature at $V_d = 10$ V from 300 K to 600 K.
2. Plot Current vs. Temperature for $V_d = 10$ V.
3. Plot Current vs. Voltage (I-V) at different temperatures ranging from 300 K to 600 K.

Mathematical Model

Under the assumption of full ionization in the extrinsic region, the free electron concentration is fixed entirely by the donor concentration ($n \approx N_D$). To ensure this assumption remains physically valid at the upper boundary of our temperature sweep (600 K), we evaluate the intrinsic carrier concentration n_i at $T = 600$ K. Using the standard relationship $n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right)$, the intrinsic concentration rises to approximately $n_i \approx 9.4 \times 10^{20}$ m^{-3}. Because our doping concentration ($N_D = 2 \times 10^{23}$ m^{-3}) remains over 200 times larger than n_i even at peak thermal excitation, thermal generation of carriers remains negligible ($N_D \gg n_i$).

Consequently, the temperature dependence of the material is governed solely by the carrier mobility μ, which degrades due to increased phonon (lattice) scattering at higher temperatures:

$$\mu = U_0 \left(\frac{T}{T_{ref}}\right)^{-1.5} \tag{1}$$

Because the carrier concentration remains constant while mobility drops, the overall conductivity ($\sigma \approx N_D \cdot q \cdot \mu$) decreases, meaning the resistance must rise as temperature increases.

Verilog-A Implementation

```
`include "constants.vams"
`include "disciplines.vams"

module res_n_doped(p, n);
    inout p, n;
    electrical p, n;

    parameter real len    = 10e-6    ;
    parameter real width  = 100e-6   ;
    parameter real thick  = 0.2e-6   ;
    parameter real U0     = 1350e-4  ;
    parameter real Nd     = 2e23     ;
    parameter real Temp   = 300.0    ; // Swept via schematic
```

```
    parameter real Tref    = 300.0     ;

    real Area, mu, sigma, R;

    analog begin
        Area = width * thick;
        mu = U0 * pow(Temp/Tref, -1.5);
        sigma = Nd * `P_Q * mu;
        R = len / (sigma * Area);
        V(p, n) <+ I(p, n) * R;
    end
endmodule
```

Listing 2: Verilog-A Code for N-Doped Silicon Resistor

Circuit Schematic

To execute the DC temperature sweep, the Verilog-A component was placed in a testbench alongside a 10 V DC voltage source and a current probe. A parameter sweep block was configured to step the global `Temp` variable from 300 K to 600 K, which dynamically updates the temperature parameter inside the Verilog-A model during the simulation.

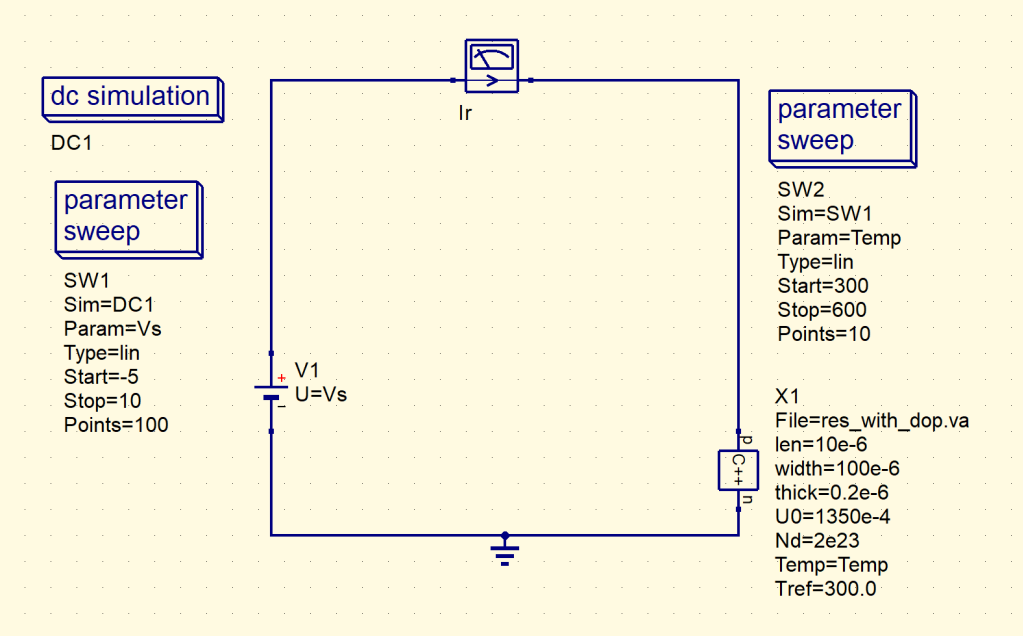


Figure 5: DC Simulation Testbench for N-Doped Silicon

Results and Deliverables

1. Resistance vs. Temperature ($V_d = 10$ V)

A DC temperature sweep from 300 K to 600 K was executed. Unlike the intrinsic case, the resistance of the n-doped wafer increased with temperature, rising from approximately 115.7 Ω at 300 K to 327.4 Ω at 600 K. Because the carrier concentration is fixed by the donor concentration, this rise in resistance is driven entirely by the degradation of carrier mobility caused by increased phonon scattering at higher temperatures.

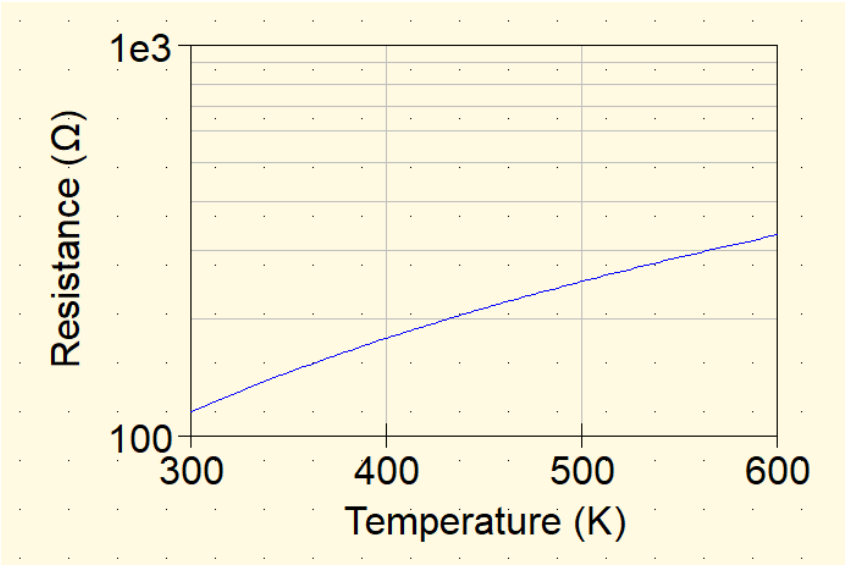


Figure 6: Resistance vs. Temperature for N-Doped Silicon

2. Current vs. Temperature ($V_d = 10$ V)

With a constant voltage of 10 V applied, the current exhibits an inverse relationship to the rising resistance. The current steadily drops from approximately 0.086 A at 300 K down to 0.03 A at 600 K, confirming the reduction in conductivity.

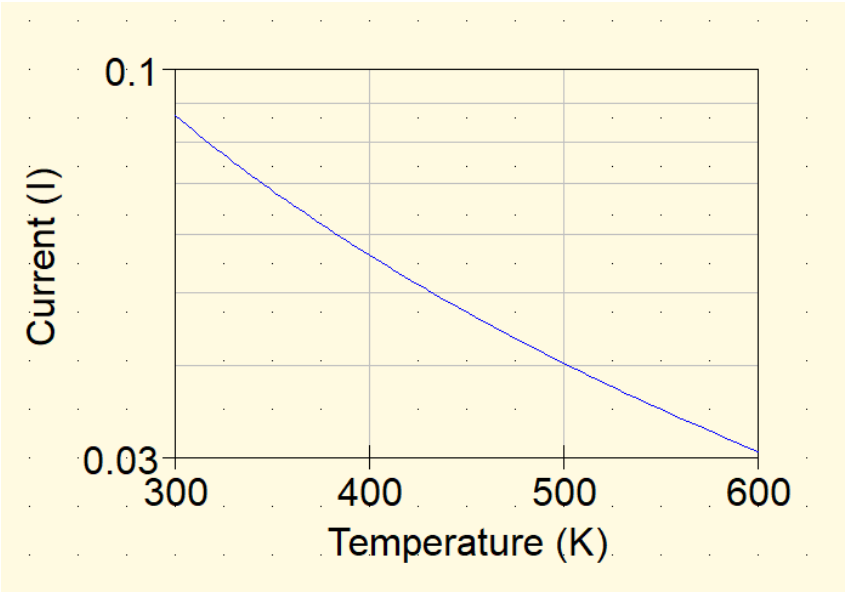


Figure 7: Current vs. Temperature for N-Doped Silicon at $V_d = 10$ V

3. Current vs. Voltage (I-V) at Different Temperatures

A family of I-V curves was generated by sweeping the voltage at various temperature steps. Because the device is an ohmic resistor, all curves remain perfectly linear. However, in direct contrast to the intrinsic silicon simulation, the slope (conductance) of the n-doped silicon decreases at higher temperatures as the lattice scattering makes the material more resistive.

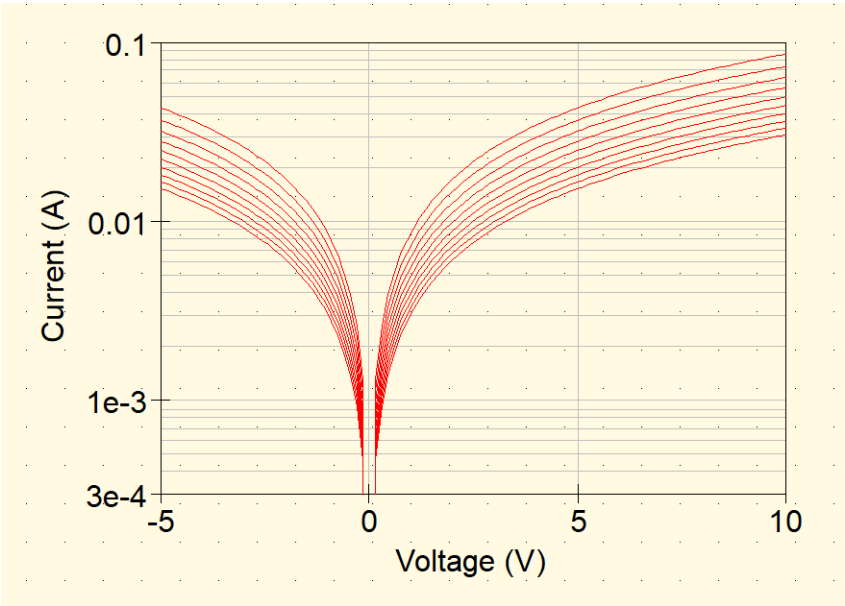


Figure 8: I-V Curves at Multiple Temperatures (300 K to 600 K) for N-Doped Silicon

Inference

The behavior of the n-doped silicon wafer is quite different from that of the intrinsic semiconductor, providing a clear picture of the physics of the extrinsic region. In the fully ionized assumption, there is a huge fixed contribution of free electrons to the carrier concentration ($n \approx N_D$). Since this value for the extrinsic region is very high, the small amount of carriers produced by the intrinsic process from 300 K to 600 K will be negligible. The dependence of the material on the temperature, having a fixed carrier concentration, will be solely due to the dependence on the carrier mobility. Since the temperature is increasing, the number of phonons due to vibration increases, resulting in more scattering processes of electrons during drift. The scattering of carriers will reduce the mobility ($\mu \propto T^{-1.5}$) and therefore conductivity.